Calculus I

Computation problems #3

Topics:

- Continuous Functions
- Derivatives

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Problem 1. The toll T charged for driving on a certain stretch of a toll road is 5\$ except during rush hours (between 7 AM and 10 AM and between 4 PM and 7 PM) when the toll is 7\$.

a) Sketch a graph of T as a function of the time t, measured in hours past midnight.

b) Discuss the discontinuities of this function and their significance to someone who uses the road.

Problem 2. Explain why each function is continuous or discontinuous.

a.) The temperature at a specific location as a function of time.

b) The temperature at a specific time as a function of the distance due south from Hanoi city.

c) The altitude above sea level as a function of the distance due east from Lao Cai city.

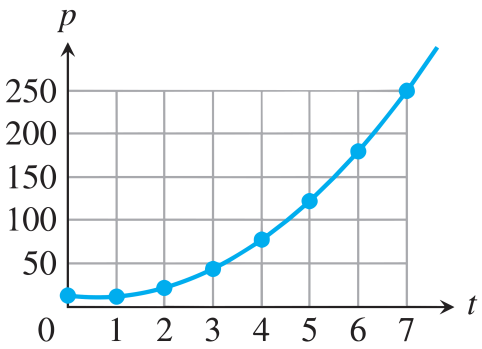
d) The cost of a taxi ride as a function of the distance traveled.

e) The current in the circuit for the lights in a room as a function of time.

Problem 3. Show that there is a root of the equation $(1-x)^3 = \sin x$ in the interval $(0, 1)$.

Problem 4. Show that the two graphs $y = e^x$ and $y = -x$ intersect.

Problem 5. *(Growth of yeast cells)* In a controlled laboratory experiment, yeast cells are grown in an automated cell culture system that counts the number P of cells present at hourly intervals. The number after t hours is shown in the accompanying figure

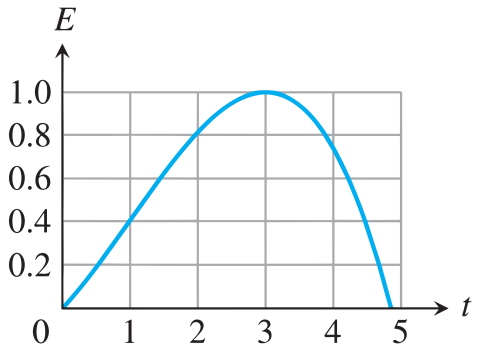


a) Explain what is meant by derivative $P'(5)$. What are its units?

b) Which is larger, $P'(2)$ or $P'(3)$? Give a reason for your answer. *Hint: Use the tangent lines.*

c) The quadratic curve capturing the trend of the data points is given by $P(t) = 6.10t^2 - 9.28t + 16.43$. Find the instantaneous rate of growth when $t = 5$ hours.

Problem 6. *(Effectiveness of a drug)* On a scale from 0 to 1, the effectiveness E of a pain-killing drug t hours after entering the blood stream is displayed in the accompanying figure.



a) At what times does the effectiveness appear to be increasing? What is true about the derivative at those times?

b) At what time would you estimate that the drug reaches its maximum effectiveness? What is true about the derivative at that time? What is true about the derivative as time increases in the 1 hour *before* your estimated time?

Problem 7.

1. For $f(s) = \dfrac{\sqrt{s} - 1}{\sqrt{s} + 1}$, find f'.

2. For $y = \dfrac{1}{(x^2-1)(x^2+x+1)}$, find y'.

3. For $g(t) = (t^2+1)\left(t+5+\dfrac{1}{t}\right)$, find g'

Problem 8.

1. Find the first and second derivatives of the function
$$r = \frac{1}{3s^2} - \frac{5}{2s}.$$

2. Find the first and second derivatives of the function
$$w = \left(\frac{1+3z}{3z}\right)(3-z).$$

Problem 9.

(a) Let $f(x)$ be a function satisfying $|f(x)| \leq x^2$ for $-1 \leq x \leq 1$. Show that f is differentiable at $x = 0$ and find $f'(0)$.

(b) Show that
$$f(x) = \begin{cases} x^2 \sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$
is differentiable at $x = 0$ and find $f'(0)$.

Problem 10. For what values of a and b is
$$g(x) = \begin{cases} ax+b, & x \leq -1, \\ ax^3+x+2b, & x > -1, \end{cases}$$
differentiable for all values of x?

Problem 11. Discuss the geometry of the resulting graph of the function g above.

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Answer to selected problems

7. (1) $\dfrac{1}{\sqrt{s}(\sqrt{s}+1)^2}$ (2) $-\dfrac{4x^3+3x^2-1}{(x^2-1)^2(x^2+x+1)^2}$ (3) $3t^2+10t+2-\dfrac{1}{t^2}$

10 . $a = -\dfrac{1}{2}$, $b = 1$.